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Privacy in Smart Homes with Remote User Authenticated Key Establishment Protocol

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Abstract

The Internet of Things (IoT) is a set of devices that contain electronic software, sensors, and actuators and exchange information using the Internet or network. Because of multiple applications, the number of IoT devices and sensors is increasing daily. Therefore, the traditional addressing security protocols cannot guarantee performance in all scenarios. The most crucial factor is the requirement of a unique addressing and secure scheme that addresses each smart IoT device or application and ensures secure communication. Therefore, robust and lightweight authentication protocols are crucial to ensuring security in smart home IoT (SH-IoT) networks. The proposed scheme provides a remote authentication protocol for the secure exchange of information between the user and the smart home IoT network. Informal and formal security analyses using the ROR model are presented in this work. The presented scheme is validated using the AVISPA tool.

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Keywords: Protocols; Session keys ; Identity verification; Key agreement; Smart Devices (SDs);Home Gateway;

1. Introduction

With the development of portable device technologies, the Internet of Things (IoT) has progressed towards ”smart homes.” The academic and scientific communities have closely monitored the progress of the IoT. IoT devices are connected to home networks either directly or through a bridge. These devices include detectors, actuators, and RFID tags. The local home gateway (HG) allows smart devices (SDs) to communicate wirelessly. SDs connect to the internet via the HG, which is an intermediary between users and SDs [1] [2]. IoT is the interconnectivity of all things to the Internet, including wearable tech and sensors that enhance smart home services, improving residents' quality of life. Smart homes can use smart devices to identify their users' health and behavioural patterns. They can also provide convenient and efficient services to people with disabilities, illnesses, and the elderly [3] [4].

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Smart home environments offer users numerous benefits, such as convenience and improved quality of life, due to the availability of new features and services. However, the data exchange over insecure channels raises serious privacy concerns, undermining the advantages of smart homes [5] [6]. Although smart homes can interact with internal and external environments daily and are becoming increasingly popular, most devices have limited capabilities, making them vulnerable to privacy breaches. Attackers could intercept data between a user's mobile device and IoT device, allowing them to determine the user's schedule [7] [8].

Smart homes, based on the IoT, have been extensively researched to provide good services to homeowners, including interoperability and resource management. However, security issues pose a significant threat to this domain. Insecure communication channels can lead to various attacks, such as man-in-the-middle (MITM) and user impersonation, which can compromise users' privacy [9]. Moreover, smart home device data can be manipulated and sold to unauthorized parties. Communication is also a challenge due to limited sensor storage and computational power. IoT platforms lack addressing and authentication schemes, leading to interoperability issues [10].

1.1. Scheme's Primary Contribution

The research proposes a secure addressing approach to distinguish users and smart equipment in the SH-IoT network.

1. The scheme monitors smart systems, gadgets, and appliances in SH-IoT with unique IDs linked to their addresses, ensuring secure and traceable transportation of personal data.
2. We used informal and formal security analysis with the ROR model and formally evaluated the proposed scheme's security with the AVISPA tool.

1.2. Related Work

Researchers have proposed authentication protocols for the secure exchange of information in smart homes. A recent paper established standards for FNC management and the IPv6 Unicast Address Format. The authors suggest using the SLAAC, or Moving Target IPv6 Defense scheme, for protection against DoS attacks [11]. They propose a low-cost, surveillance, DoS attack-resistant IPv6 hopping protocol for IoT networks. Additionally, the authors covered various strategies using a global unicast address format, which the author cited in [12] [13], restoring it in various ways. In IoT, the global unicast IPv6 address poses security risks due to the large number of smart devices. Cryptographic two-factor security mechanisms have been proposed, but single-server implementations are unsuitable for distributed frameworks with multiple servers. A novel multi-server plan that uses key agreement authentication and an ECC Diffie-Hellman key scheme has been proposed. However, this scheme is vulnerable to MIMA attacks. The paper also discusses privacy and traceability attacks where the attacker can obtain the device's location and the static mode storage of the identity stored in their hardware memory [14] [15]

2. Proposed System Model

The proposed system, as shown in Figure 1, has three parts: users, RA (Registration Authority), and SHG (Smart home gateway). Users access services via smart devices, which are authenticated by the RA and assigned unique addresses. Encrypted identities are stored at SHG, and users match the MIID bit to access home appliances.

2.1. System Model

The system has three components: a smart homeowner (User), a home gateway (HG), and a control unit. The HG has smart devices and a monitoring device. All devices connect to the internet through IEEE 802.11 and are addressed using IPv6. The central control unit assigns unique addresses and identifiers to users and devices, and new user registration requires HG verification. Encrypted identities of owners are stored at the SHG [11].

2.2. Attacker Model

The Dolev-Yao (DY) threat model is used to analyze the robustness of our protocol and is described in this section. The proficiency of attacker $\mathcal{J}$ follows.

1. Attacker $\mathcal{J}$ can obstruct, inject, replay, and alter information sent over an insecure medium.
2. Using the power analysis attack, attacker $\mathcal{J}$ can get access to the legit user's device and get the secret credentials.
3. Attacker $\mathcal{J}$ has access to each party's short-term and long-term keys as well as session states [16].

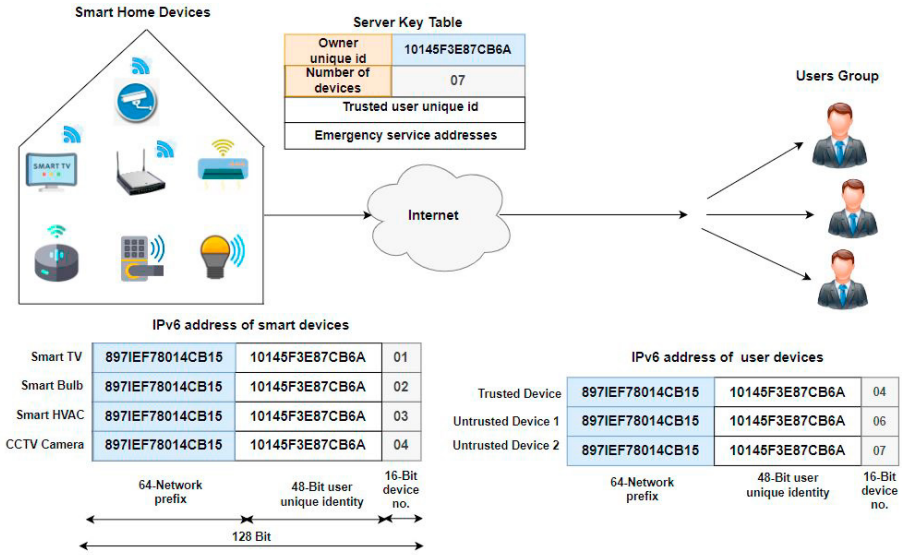


Fig. 1: System model and address format for devices and applications

3. Secured Authenticated Protocol for Smart Home (SAPSH)

SAPSH provides unique identities to user devices and smart appliances, allowing encrypted access. Our secure authentication protocol addresses IoT-based smart home network security issues in four steps.

3.1. Initialization Phase

The smart home network includes monitoring devices and smart appliances, each with a unique identity stored in the home gateway (HG). The user's encrypted identity and smart device network prefix are stored in HG. Reg creates a master key K_{Reg} using their secret keys K_a and unique IDs ID_{hg} and ID_a.

3.1.1. User registration Phase

User (U) must register at *Reg* before accessing information from device A. The following steps are involved in the registration process:

1. Device A generates a random nonce N_a and computes F_a using the hash function. Then, A sends $\{F_a, N_a\}$ to *Reg* through a secure channel.

Table 1: List of Symbols and Descriptions

Symbols	Descriptions
$MIId_u$	48-bit modified user interface identity
$MIID_a$	16-bit smart device interface identity
ID_{hg}	Identity of home gateway
K_{Reg}	Master key of *Reg*
K_a	Master secret key of smart device
PW_u	Password identity of the user
K_{uhg}	shared key between U and HG
K_{hga}	shared key between A and HG
S_k	a random number by U
h(.)	hash function (One-way)
$\|\|$	the symbol used for Concatenation
T	Time stamp
$\oplus$	the symbol used for xor
E_k	encrypting M using k
D_k	decrypting M using k

2. Upon receiving the message, *Reg* generates another random nonce N_{Reg} and computes K_{hga} using the hash function. Then, *Reg* stores $\{F_a, K_{hga}$, and $N_a\}$ in its database and sends $\{K_{hga}\}$ to A through a secure channel. *Reg* also makes F_a public.
3. After receiving the message from *Reg*, device A computes X_1, X_2, and X_3.
4. The user chooses an ID and password $\{ID_u, N_u\}$ and generates a random number N_u. Then, User(U) computes F_u using the hash function and sends $\{F_u\}$ to *Reg* through a secure channel.
5. Upon receiving the message, *Reg* computes K_{uhg} and RID_u using the hash function. Then, *Reg* stores $\{F_u, RID_u, K_{hga}\}$ in its database and sends $\{K_{hga}, RID_u\}$ to U through a secure channel. Finally, User(U) computes Y_1, Y_2, Y_3, Y_4, and F_u and stores them on the mobile device.

3.2. Authentication and Key establishment Phase

1. To authenticate, the user provides identity and password $MIID_u$, PW_u. User (U) computes N_u, HPW_u, and Y_2^*. The user(U) checks if Y_2^* is equal to Y_2, then generates N_m and computes RID_u, K_{uhg}, V_1, and W_u, and sends F_u, V_1, S_1, and W_u to HG.
2. HG processes hashes and values, checks for equality, generates a nonce, calculates more hashes, and sends the result to device A.
3. Device A generates two additional values if the computed value matches the expected value, which is sent to HG.
4. HG computes different hash values and checks them against each other to ensure the message's validity. Finally, HG sends two values to the user (U).
5. User (U) computes, updates, and replaces hashes and values for IDs and keys. It then computes and sends a hash to HG.
6. After receiving a message from User (U), HG computes V_6^* and checks if it matches V_6. If it does, delete $\{F_u, RID\}$ in the database.

3.3. Password update phase

To change your password, follow these steps:

1. Enter $MIID_u$ and PW_u^{old} on the smart device.
2. The device will calculate N_u, HPW_u^*, and Y_2 and check if $Y_2^* = Y_2$.

Table 2: Login and key establishment phase

Mobile User	**Home Gateway (HG)**	**Smart Device**
$\{Y_1, Y_2, Y_3, Y_4, F_u\}$ **Step 1** :- Inputs $MIID_u, PW_u$ Computes $N_U = Y_1 \oplus hash(MIID_u \| PW_u)$ $HPW_u = hash(MIID_u \| PW_u)$ $Y_2^* = hash(MIID_u \| PW_u \| N_u \| HPW_u)$ Checks $Y_2^* = Y_2$ Generates N_m Computes $RID_u = Y_3 \oplus hash(N_u \| HPW_u)$ $K_{uhg} = Y_4 \oplus hash(RID_u \| HPW_u)$ $V_1 = hash(F_u \| RID_u \| K_{uhg}) \oplus (N_m \| F_a)$ $S_1 = hash(MIID_u \| N_m) \oplus hash(K_{uhg} \| N_m)$ $W_u = hash(F_u \| RID_u \| N_m \| F_u \| k_{uhg})$ $\rightarrow$ $\{F_u, V_1, S_1, W_u\}$	$\{(F_u, RID_u, K_{uhg}), (F_u, K_{hga}, N_a), ID_{hg})\}$ **Step 2**:- Retrieves RID_U and K_{uhg} corresponds to F_u Computes $(N_m^* \| F_u^*) = V_1 \oplus hash(F_u \| RID_u \| N_m^* \| K_{uhg})$ $W_u^* = hash(F_u \| RID_u \| N_m^* \| K_{uhg})$ Checks $W_u^* = W_u$ Generates N_{hg} Computes $V_2 = hash(N_m \| N_{hg})$ $V_3 = hash(F_u \| K_{hga} \| N_a) \oplus V_2$ hash= $(MIId_u \| N_m) = S_1 \| hash(K_{uhg} \| N_m$ $S_2 = (hash(MIID_u \| N_m) \| (ID_{hg} \| N_{hg})) \oplus hash(K_{hga} \| N_a)$ $W_{uhg} = hash \leftarrow$ $\{F_u, V_3, S_2, W_{uhg}\}$	**Step 3**:- Computes $N_a = X_1 \oplus hash(MIID_a \| K_a)$ $K_{hgh} = X_2 \oplus hash(N_a \| K_a)$ $V_2^* = V_3 \oplus hash(F_a \| K_{hga} \| (N_a)$ $W_{uhg} = hash(F_a \| V_2^* \| K_{hga})$ Checks W_{uhg}^* Generates N_a Computes $(hash(MIID_a \| N_m)) \| (ID_{hg} \| N_{hg}))$ $S = S_2 \oplus (hash(MIID_u \| N_m)$ SK= $hash(hash(MIID_u \| N_m) \| hash(MIID_a \| N_a)$ $W_a = hash(F_u \| F_a \| V_2^* \| hash(ID_a \| N_a) \| K_{hga})$
Step 5: Computes $F_U^n = hash(F_u \| N_m)$ $hash(Id_{hg} \| N_{hg}) \| hash(MIID_a \| N_a) \| F_U^n = V_5 \oplus hash(RID_u \| N_m)$ $W_{hga}^* = hash(F_u \| N_m \| hash(Id_{hg} \| N_{hg})$ Checks $W_{hga}^* = W_{hga}$ Computes SK= $hash(hash(MIID_u \| N_m) \| hash(ID_a \| N_a)$ Updates $RID_u^n = hash(F_U^n \| K_{uhg})$ $Y_3^n = RID_u^n \oplus hash(N_u \| HPW_u)$ $Y_4^n = K_{uhg} \oplus hash(RID_u^n \| HPW_U)$ Replaces $\{Y_3, Y_4, F_u\}$ to $\{Y_3^n, Y_4^n, F_u^n\}$ in the mobile device, Computes $V_6 = hash(S_k \| F_u^n)$ $\rightarrow$ $\{V_6\}$	**Step 4**: Computes $hash(ID_a \| N_A) = V_4 \oplus hash(F_a \| K_{hga}$ $W_a^* = hash(F_u \| F_a \| V_2 \| hash(MIID_a \| N_A) \| K_{hga}$ Checks $W_a^* = W_a$ Computes $S_k = hash(hash(MIID_u \| N_m) \| hash(ID_{hg} \| N_{hg}) \| MIID_a \| N_a)$ $F_U^n = hash(F_u \| N_m)$ $RID_U^n = hash(F_U^n \| K_{uhg})$ $V_5 = hash(RID_u \| N_m) \oplus hash(ID_{hg} \| N_{hg}) \| hash(MIID_a \| N_a) \| F_U^n)$ $W_{hga} = hash(F_u \| N_m \| hash(ID_a \| N_A) \| F_U^n \| K_{uhg})$	
	Step 6: $V_6^* = hash(S_K \| F_U^n)$ Checks $V_6^* = V_6$ If it is correct then delete $\{F_u, RID_u\}$ in the database	

3. Enter the new password, PW_u^{new}.
4. The device will compute a hash and replace certain values with their hashed counterparts.

4. FORMAL SECURITY ANALYSIS

This section analyzes the SAPSH protocol using AVISPA and the Random or Real (ROR) model. The protocol involves three entities for authentication and key establishment: the mobile user, the home gateway attacker, and the smart device.

- Execute ($\mathcal{P}_U^{tp1}, \mathcal{P}_U^{tp2}, HG, \mathcal{P}_U^{tp3}, A$): Attacker $\mathcal{J}$ runs to receive messages sent between them over a public channel U, HG, and A.
- CorruptMD ($\mathcal{P}_U^{tp1}$): This query represents the attacker's ability to extract sensitive information from the user's mobile device.
- Reveal($\mathcal{P}_U^{tp}$): This query reveals the current session key SK between $\mathcal{P}_U^{tp1}$) and ($\mathcal{P}_U^{tp3}$) SD. If the Reveal ($\mathcal{P}_U^{tp}$) query cannot disclose SK between ($\mathcal{P}_U^{tp1}$) and ($\mathcal{P}_U^{tp3}$), then SK is secure.
- Send ($\mathcal{P}_U^{tp}, V$): It allows attacker $\mathcal{J}$ to send information V to ($\mathcal{P}_U^{tp}$) and receive a response.
- Test ($\mathcal{P}_U^{tp}$): A fair coin fc is flipped. If the session key SK is valid, ($\mathcal{P}_U^{tp}$) returns SK for $fc = 1$ and a random number for $fc = 0$. If SK is invalid, ($\mathcal{P}_U^{tp}$) returns an empty string.
- A must differentiate the outcome value from the Test query on ($\mathcal{P}_U^{tp}$). Attacker $\mathcal{J}$ checks the consistency of the random bit fc using the Test query output. If fc equals 0, then $\mathcal{J}$ wins.
- **Theorem 1**. Our approach's semantic protection is demonstrated by the ROR model, which uses Zipf's law and uniform distribution to ensure security despite password size constraints.

Let's consider an $\mathcal{J}$ attacker operating in the ROR model's upper bound polynomial time T_n. Consequently, the probability of a security breach is given by the equation:

```
role user (MU, SD, HG, RA: agent, SKmura,
SKsdra: symmetric_ key, H: hash func, SND,
RCV: channel (dy))
Played_by MU
Def=
Local State: nat.
IDmu, PWmu, Rmu, PIDmu, Kmug, Kra, Rra,
RIDmu, HPWmu, A1, A2, A3, A4, IDsd, Rsd,
PIDsd, Kgsd, B1, B2, Ksd: text, RNmu, M1, C1,
Vmu, IDg, RNg. M2, M3, C2. Vmug. RNsd, SK,
M4, Vsd, M5, Vgsd, PIDmunew, RIDmunew,
A3new, A4 new, M6: text
const spl, sp2, sp3, sp4, mu_hg_mmu, hgw_sd
rug, sd hgw rusd, hgw mu rag, hgw mu rusd:
protocol id init State := 0 transition
%%%%%%%%%%%Registration phase
1. State = 0 /\ RCV (start) =|>
State' = 1 /\Rmu' = new () /\ PIDmu':= H
(IDmu. Rmu')
/\ SND ({PIDmu'} _SKmura)
/\ secret ( {IDmu, PWmu), sp1, {MU})
2. State = 1 /\ RCV (H {H (IDmu. Rmu') Kra.Rra)
H(H(IDmu.Rmu') H(H(IDmu.Rmu'). Kra.Rra))
Rsd'} _SKmura)-1> State':= 2 /\ HPWmu':=
H(PWmu Rmu')
/\ A1':= xor (Rmu',H(IDmu PWmu)) /\ A2':-
H(IDmu. PWmu.Rmu' HPWmu)
/\ A3' =
xor(H(H(IDmu.Rmu').H(H(IDmu.Rmu').Kra.Rra)
),H(Rmu 'HPWmu))
/\ A4':= xor (H(H(IDmu.Rmu').Kra.Rra), H
(H(H(IDmu.Rmu').
H(H(IDmu.Rmu').Kra.Rra)).HPWmu'))%%%%
%%%%%%%%%Authentication & Key
agreement phase
/\RNmu':= new ()
/\ MI':= xor (H(H(IDmu.Rmul H (H
(IDmu.Rmu).H(H(IDmu.Rmu'). Kra.Rra)).
H(H(IDmu.Rmu'). Kra.Rra)), H (RNmu' H
(IDsd. Rsd')])
/\ C1':= xor(H(IDmu RNmu'),H(H(H(IDmu.
Rmu') Kra.Rra) RNmu'))
/\ Vmu' := H(H(IDmu.Rmu'),H(H(Imu.Rmu').
H(H(IDmu Rmu'). Kra.Rra)) RNmu' H
(IDsd.Rsd) H (H(IDmu Rmu') Kra.Rra))
/\ SND (H(IDmu.Rmu'). MI'.CI' Vmu')
/\ witness (MU, GWN, mu hg _rnmu, RNmu')
3. State = 2 /\
RCV(xor(H(H(H(ImuRmu').H(H(ImuRmu').Kra.
Rra)).RNmu').(H(Ig.RNg').H(IDsd.RNsd')H(H(I
Dmu.Rmu')
RNmu'))) H(H(IDmu Rmu').RNmu'.
H(IDg.RNg'). H(IDsd.RNsd').H(H(IDmu.Rmu')
_Kra.Rra))) -b
State':= 3 A SK':- H(H(IDmu.
RNmu').H(IDg.RNg').H(IDsd.RNsd'))
/\ M6':- H (SK'. H(H(Dmu Rm').RNmu').
/\ SND(M6')
/\ request(GWN, MU,hg_mu_mg.RNg)
/\ request (GWN.MUhgw_mu_rsd, RNsd')
end role
```

Fig. 2: Role for User/homegateway

```
role session(MU, SD, HG, RA: agent, SKmura, SKsdra:
symmetric key, H: hash func)
def=
local SN1, SN2, SN3, SN4, RV 1, RV2, RV3, RV4 : channel
(dy)
composition
user (MU, SD, HGW, RA, SKmura, SKsdra, H, SNI, RV1)
/\ device (MU, SD, HG, RA, SKmura, SKsdra, H, SN2,
RV2)
/\ gate(MU, SD, HG, RA, SKmura, SKsdra, H, SN3, RV3)
/\ register(MU, SD, HG, RA, SKmura, SKsdra, H, SN4,
RV4)
end role
role environment)
def=
const mu, sd, hg, ra: agent, skmura, sksdra: symmetric
key,
h: hash func,
idmu, idsd, idg, kra, ksd, kmug, kgsd, pidmu, pidsd: text,
mu_hg_rnmu, hg _sd_rng, sd_hg _rnsd, hg_mu _Rng,
hg _mu_rnsd: protocol id, sp1, sp2, sp3, sp4: protocol id
intruder_knowledge = {mu, sd, hgw, ra, pidmu, pidsd,
h}
composition
/\session(mu, sd, hg, ra, skmura, sksdra, h) /\session(i,
sd. hg, ra, skmura, sksdra, h)
/\session(mu, i, hg, ra, skmura, sksdra, b) /\ session(mu,
sd, i, ra, skmura, sksdra, h)
/\session(mu, sd, hg, i, skmura, sksdra, h)
end role
security goals
secrecy_of sp1, sp2, sp3, sp4
authentication_on mu_hg_rmu
authentication_on hg_sd_rng
authentication_ on sd_ hg_rnsd
authentication on hg _mu_rng
authentication on hg_mu_rnsd
end goal
environment()
```

Fig. 3: Role for Session/environment

% OFMC % Version of 2006/02/13 SUMMARY **SAFE** DETAILS BOUNDED_NUMBER_OF_SESSIONS PROTOCOL /home/span/span/testsuite/results/ SAPSH.if GOAL as specified BACKEND **OFMC** COMMENTS STATISTICS parseTime: 0.00s searchTime: 0.44s visitedNodes: 0 nodes depth: 4 plies	% CL-AtSe SUMMARY **SAFE** **CL-AtSe** DETAILS BOUNDED_NUMBER_OF _SESSIONS PROTOCOL TYPED MODEL PROTOCOL /home/span/span/testsuite/results/ SAPSH.if GOAL As specified BACKEND **CL-AtSe** STATISTICS Analysed: 0 states Reachable: 0 states Translation: 0.04 seconds Computation: 0.02 seconds

Fig. 4: AVISPA result using OFMC and CL-AtSe

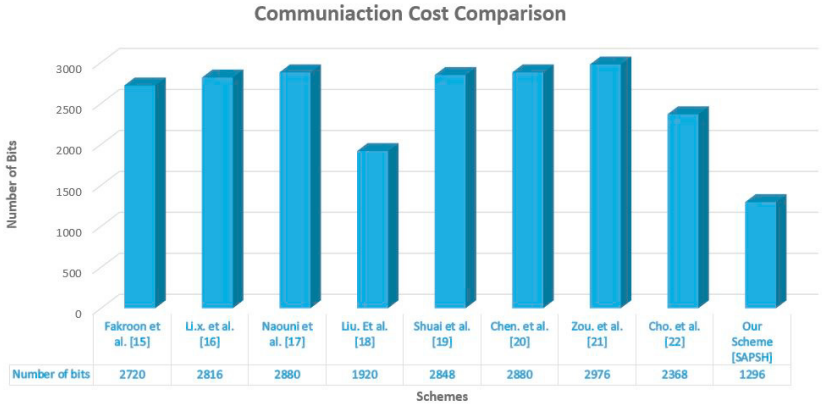


Fig. 5: Comparison of Communication cost

$Adv_{\mathcal{J}}^{SAPSH} \leq \frac{query_h^2}{|h(.)|} + \frac{query_h}{|D|}$· where $query_h$ h denotes the total number of hash queries $_{send}$: represents the query being sent, h(.) represents the hash function, and $|D|$ represents the dictionary's size to identify the password.

Proof: Let GM_i represent the set of events where i is in the range (1 to 4) and where this event is used as the likelihood for assuming bit c in a test simulation that starts with GM_0 and ends with GM_4.

GM_0: By choosing and guessing the bit c at the beginning of the game, the attacker $\mathcal{J}$ executes the attack on the proposed schemes. Next, security is added to our method by:

$$Adv_{SAPSH}^{J} = 2Pr[GM_0] - 1. \tag{1}$$

GM_1: Attacker $\mathcal{J}$ seeks session and secret key and verifies authenticity by revealing a test query matched against a random key. Packets are timestamped to prevent eavesdropping.

$$Pr[GM_0] = Pr[GM_1]. \tag{2}$$

GM_2: The attacker $\mathcal{J}$ initiates a computation inquiry via hash and sends fake queries to calculate SK and two other values, remaining undetected.

$$Pr[GM_1] - Pr[GM_2] \leq \frac{query_h^2}{|h(.)|} \tag{3}$$

GM_3: GM_2 Fake queries that reduce smart device usage can lead to GM_3 access. Dictionary attacks are often used to guess passwords, but their effectiveness is limited by reduced entropy and hash collisions.

$$Pr[GM_2] - Pr[GM_3] \leq \frac{query_{send}}{2^n.|h(.)|} \tag{4}$$

GM_4: Attacker $\mathcal{J}$ uses online and offline attacks to acquire HG, user sessions, and secret keys. Corrupt simulation queries and a hash random oracle are used for offline attacks. However, identifying GM_3 or GM_4 requires password-guessing.

$$Pr[GM_3] - Pr[GM_4] \leq \frac{query_{send}h}{2.|h(.)|} \tag{5}$$

GM_4 simulations showed that the suggested protocol is secure against the attacker $\mathcal{J}$. Despite using the $Test()$ simulation query to launch an attack, the attacker was unsuccessful. Therefore, the likelihood of GM_3 is:

$$Pr[GM_4] \leq \frac{1}{2} \tag{6}$$

$$\frac{1}{2}Advr_{\mathcal{J}}^{SAPSH} \leq \frac{q^2h}{|hash|} + \frac{2qsend}{|D|}. \tag{7}$$

Using Equations (1-6), we determined password attack parameters, and the SAPSH scheme prevents password and session fundamental assumptions by attackers.

4.1. Simulation using AVISPA Tool

AVISPA is security analyzer software with a single button that automates the validation of critical Internet protocols and applications. This paper analyzes the security of the SAPSH approach against passive and active attacks using the AVISPA tool. We only consider the OFMC and CL-ATSe back-end analysis methods. The simulation results show that the protocol achieves its security objectives, as confirmed by Figures 2 and 3, and supported by Figure 4.

5. Performance evaluation

Our study compared our protocol to other existing authentication methods regarding functionality, communication, and computation costs. Our results show that SAPSH outperforms other methods regarding performance, communi-

Table 3: Comparison of Computation cost

Schemes	Total Computation cost	Execution Time(milliseconds)	No. of bits	No. of messages
Fakroon et al. [17]	$25T_h + 1T_R + 4T_s$	0.0312	2720	4
Li.x. et al. [18]	$11T_h + 4T_s$	8.0314	2816	4
Naouni. et al. [19]	$3T_{mul} + 6T_s + 17\ T_h$	6.0307	2880	4
Shuai .et al.[20]	$16T_h + 3T_m$	6.0086	1920	4
Liu. et al. [21]	$26T_h + 1T_b$	14.749	2848	4
Chen. et al. [22]	$33T_h$	6.1650	2880	5
Zou. et al. [23]	$6T_{mul} + 6T_h$	11.9496	2976	4
Cho. et al .[24]	$34T_h + T_p + T_f$	2.197	2368	4
Our Scheme (SAPSH)	$42T_h + 2_E + 2_D$	0.164	1296	4

Table 4: Comparison Functionality and security attribute

Attacks	**[17]**	**[18]**	**[19]**	**[20]**	**[21]**	**[22]**	**[23]**	**[24]**	**SAPSH**
MUI	✓	×	✓	✓	✓	✓	✓	✓	✓
HGI	×	✓	×	✓	×	✓	✓	✓	✓
SDI	✓	✓	✓	−	✓	✓	✓	✓	✓
SKD	✓	✓	✓	−	✓	✓	✓	✓	✓
RM	✓	✓	×	✓	✓	−	✓	−	✓
AU	✓	✓	✓	✓	✓	✓	✓	✓	✓
OG		✓	✓	×	✓	✓	−	✓	✓
SAPSH	✓	✓	✓	✓	✓	✓	✓	✓	✓

Note 1: ✓ ; the plan supports a specific security feature or functionality. Note 2: × ; The plan is insecure or does not provide specific functionality. Note 3: − ; Not applicable in the functionality or security feature.

cation, and computation cost. The home gateway verifies the user's request before assigning a unique identity, which allows them to log in using an ID and password.

5.1. Functionality Comparison

This section presents the scheme's features and safeguards against attacks. These include attacks on mobile users, home gateways, smart device impersonation, session key disclosure, replay, MITM, anonymity, untraceability, offline guessing attacks, and more. The presented scheme facilitates dynamic session key establishment, as demonstrated in Tables 2 and Table 3. The comparative analysis indicates that the proposed scheme can authenticate users and IoT devices successfully.

5.2. Communication Cost and Computational Cost

To demonstrate the efficiency of our proposed authentication scheme, we have evaluated its computation cost and compared it with another authentication scheme. The comparison revealed that our authentication protocol has a lower computational cost. As shown in table 2, T_m, T_R, T_h, and T_s represent the time it takes for symmetric key encryption/decryption (0.1303 ms), a hash function (0.0004 ms), a point multiplication using ECC (7.3529 ms), and an extractor with fuzzy logic, respectively, which only makes use of the one-way hash function. Our proposed protocol, the SAPSH, has better security and performance than existing schemes. The authentication and key agreement phase requires a total communication cost of 1296 bits, as shown in Table3 and Figure 5.

6. Conclusion

The smart homes of the future will use IoT networks for remote access. To ensure secure communication, authentication is a must. The SAPSH scheme offers a unique way to authenticate smart devices and appliances. Smart home networks allow remote control of appliances using personal devices. Addressing security concerns is crucial for smart homes to be integrated into society. The SAPSH provides secure and lightweight protocols that meet verification and message integrity requirements while minimizing risks. The security verification results of the proposed approach are promising, making it suitable for real-world IoT-SH applications.

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